MAIN-SEQUENCE STARS

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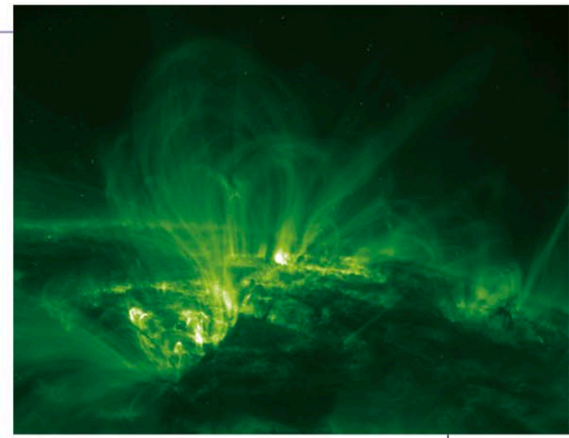
MAIN-SEQUENCE STARS are those that convert hydrogen into helium in their cores by nuclear reactions. Stars spend a high proportion of their lives on the main sequence, during which time they are very stable. The higher the mass of the star, the less time it spends on the main sequence, as nuclear reactions occur faster in higher-mass stars.



STAR FLARES
The Sun's photosphere radiates huge amounts of energy as solar flares contribute to the solar wind.

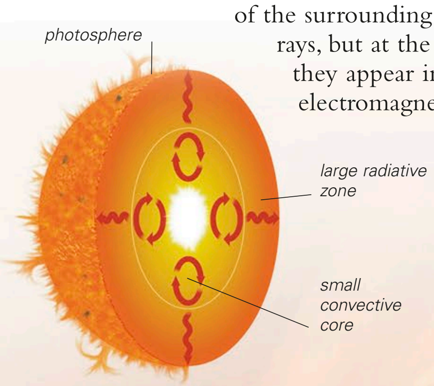
STAR ENERGY

The cores of main-sequence stars initially consist mainly of hydrogen. When the temperature and pressure become high enough, the hydrogen is converted into helium by nuclear reactions. For stars of less than about 1.5 solar masses, this is done by means of a process called the proton–proton chain reaction (the pp chain). For stars of more than about 1.5 solar masses and with core temperatures of more than about 36 million °F (65 million °C), carbon, nitrogen, and oxygen are used as catalysts in a process called the carbon cycle (CNO cycle). When hydrogen is converted to helium, a tiny amount of energy is released as gamma rays, which gradually permeate their way out through the photosphere (the Sun's visible surface). The huge amounts of energy radiated by main-sequence stars are due to the immense masses of hydrogen they contain. In the core of the Sun, 600 million tons of hydrogen are converted into helium every second.

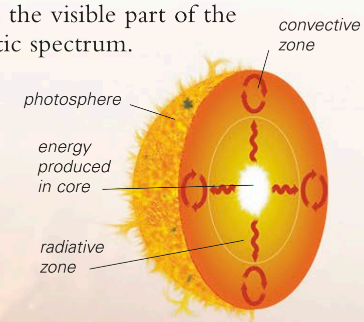
MASSIVE STAR
Achernar, or Alpha (α) Eridani, the ninth-brightest star in the sky, is a blue main-sequence star of about 6 to 8 solar masses. Main-sequence stars of this size convert hydrogen to helium through a process called the carbon cycle.

STELLAR STRUCTURE

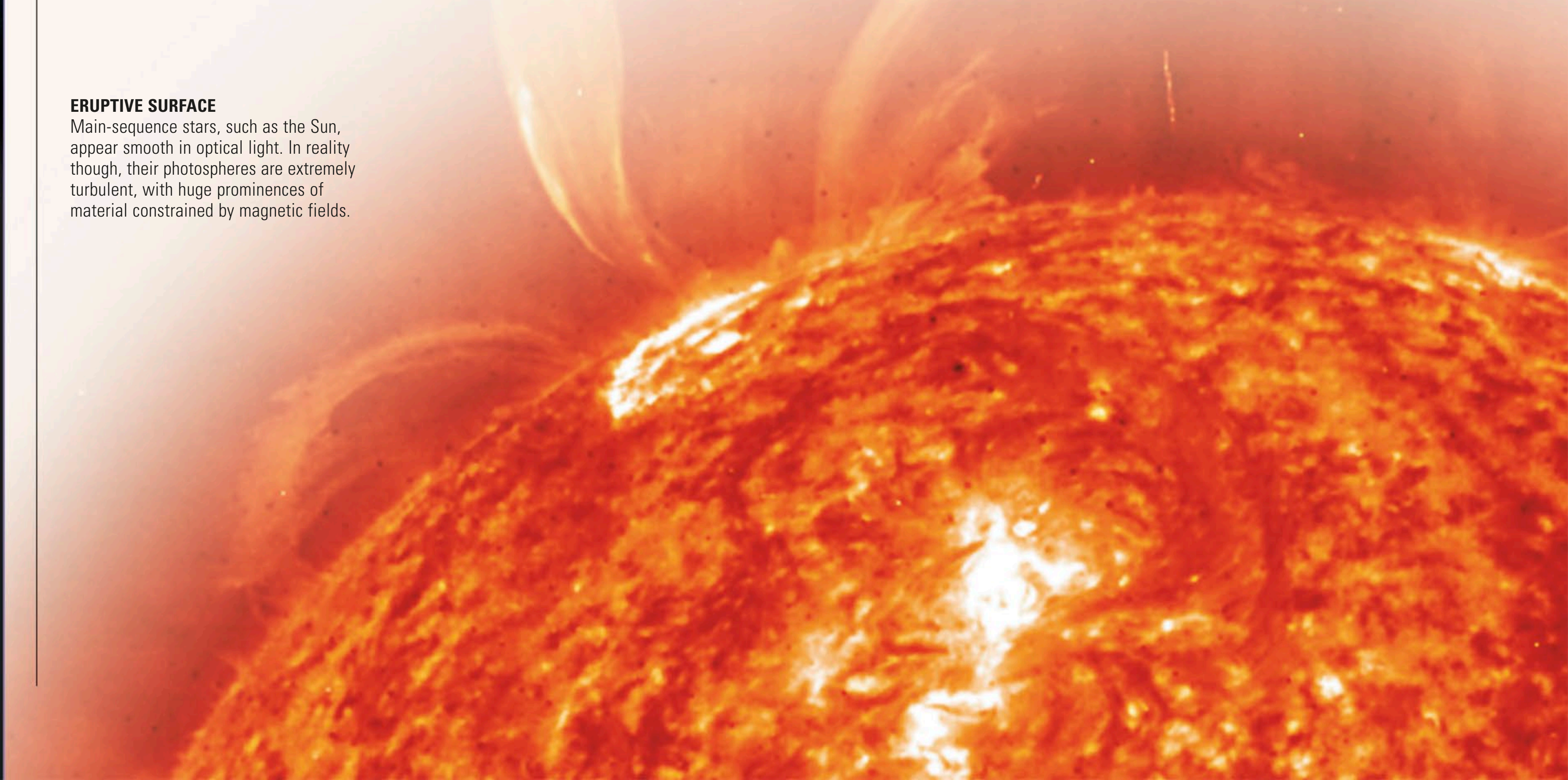
Energy, in the form of gamma rays, is released in the nuclear reactions occurring within stellar cores. This energy can be transported outward by two processes: convection and radiation. In convection, hot material rises to cooler zones, expanding and cooling, then sinks back to hotter levels, just like water being boiled in a saucepan. In the radiation process, photons are continually absorbed and reemitted. They can be emitted in any direction, and sometimes travel back into the central core. They follow a path called a "random walk" but gradually diffuse outward, losing energy as they do so. Their energy matches the temperature of the surrounding material, so they start as gamma rays, but at the Sun's surface (the photosphere), they appear in the visible part of the electromagnetic spectrum.



HIGH-MASS STAR
Stars with a mass greater than 1.5 solar masses produce energy through the CNO cycle. They have convective cores and a large radiative zone reaching to the photosphere.



LOW-MASS STAR
In stars with a mass smaller than 1.5 solar masses, the pp chain dominates and a large, inner radiative zone reaches out to a smaller convection zone near the star's photosphere.



ERUPTIVE SURFACE
Main-sequence stars, such as the Sun, appear smooth in optical light. In reality though, their photospheres are extremely turbulent, with huge prominences of material constrained by magnetic fields.